SQL Constraints with CREATE TABLE Examples

A constraint is a rule applied to a table or column to ensure that only valid, accurate, and consistent data is stored.

Constraint	Main purpose
PRIMARY KEY	Uniquely identifies each row
FOREIGN KEY	Creates and enforces a relationship between tables
UNIQUE	Prevents duplicate values
NOT NULL	Prevents missing values
CHECK	Allows only values that satisfy a condition
DEFAULT	Adds a value automatically when none is supplied

1. PRIMARY KEY

A PRIMARY KEY uniquely identifies every row in a table. It does not allow duplicate values or NULL values.

```
CREATE TABLE Student (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(100),
    Age INT
);
```

2. FOREIGN KEY

A FOREIGN KEY creates a relationship between two tables. The value stored in the child table must already exist in the parent table.

```
CREATE TABLE Department (
    DepartmentID INT PRIMARY KEY,
    DepartmentName VARCHAR(100)
);

CREATE TABLE Employee (
    EmployeeID INT PRIMARY KEY,
    EmployeeName VARCHAR(100),
    DepartmentID INT,
    FOREIGN KEY (DepartmentID)
        REFERENCES Department(DepartmentID)
);
```

3. UNIQUE

A UNIQUE constraint prevents duplicate values in a column. It is commonly used for email addresses, phone numbers, or employee codes.

```
CREATE TABLE Customer (
    CustomerID INT PRIMARY KEY,
    CustomerName VARCHAR(100),
    Email VARCHAR(150) UNIQUE
);
```

4. NOT NULL

A NOT NULL constraint makes a column mandatory. A row cannot be inserted without providing a value for that column.

```
CREATE TABLE Product (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(100) NOT NULL,
    Price DECIMAL(10,2)
);
```

5. CHECK

A CHECK constraint allows data only when it satisfies a specified condition.

```
CREATE TABLE EmployeeSalary (
    EmployeeID INT PRIMARY KEY,
    EmployeeName VARCHAR(100),
    Age INT CHECK (Age >= 18),
    Salary DECIMAL(10,2) CHECK (Salary > 0)
);
```

6. DEFAULT

A DEFAULT constraint automatically inserts a predefined value when the user does not provide one.

```
CREATE TABLE Orders (
    OrderID INT PRIMARY KEY,
    CustomerName VARCHAR(100) NOT NULL,
    OrderStatus VARCHAR(30) DEFAULT 'Pending',
    OrderDate DATE DEFAULT GETDATE()
);
```

Combined Example

The following table uses several constraints together in one CREATE TABLE statement.

```
CREATE TABLE Employee (
    EmployeeID INT PRIMARY KEY,
    EmployeeName VARCHAR(100) NOT NULL,
    Email VARCHAR(150) UNIQUE,
    Age INT CHECK (Age >= 18),
    Salary DECIMAL(10,2) CHECK (Salary > 0),
    Country VARCHAR(50) DEFAULT 'India',
    DepartmentID INT,
    FOREIGN KEY (DepartmentID)
        REFERENCES Department(DepartmentID)
);
```

Quick Memory Guide

Constraint	Remember it as
PRIMARY KEY	Unique row identifier
FOREIGN KEY	Link between tables
UNIQUE	No duplicate values
NOT NULL	Value is compulsory
CHECK	Value must satisfy a condition
DEFAULT	Automatic value when omitted